

# Roots of $d$ -matching polynomials and the spectrum of the universal cover: a closed form at first Betti number one, and a refuted conjecture

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## Abstract

Hall, Puder and Sawin place the roots of the  $d$ -matching polynomial  $\mu_{d,G}$  in the interval  $[-\rho, \rho]$ ,  $\rho$  the spectral radius of the universal cover  $T$ . We conjectured that they lie in  $\text{spec}(T)$  itself, a proper subset of that interval whenever  $T$  has a spectral gap: no root ever enters a gap. This strengthens their theorem and is the matching-polynomial form of their Question 6.3.

We prove it for every connected  $G$  of first Betti number one and every  $d$ , by giving  $\mu_{d,G}$  in closed form.

In general it is false. The counterexample is due to Chris Hall [12]: a bipartite graph on 41 vertices with  $\mu_G = x^{21}(x^4 - 11x^2 + 25)^4(x^2 - 5)(x^2 - 11)$ , where  $\sqrt{5}$  lies in an internal gap of  $\text{spec}(T)$ .

The natural repairs, minimum degree three and 3-connectivity, are open. Our main contribution is a structural reason why counterexamples are hard to build, and why the known one looks as it does. If  $\delta(G) \geq 3$  and  $\theta \neq 0$  is an eigenvalue of  $T_G$ , then

$$\theta^2 > \frac{8(\delta - 2)^2}{(\Delta - 2)(\Delta + 2\delta - 6)}, \quad \text{so} \quad |\theta| > \frac{2\sqrt{2}(\delta - 2)}{\Delta}(1 + o(1))$$

(Theorem 21): an eigenvalue near zero forces degree spread growing like  $1/|\theta|$ . The bound is attained for every admissible pair of parameters (Theorem 22), the smallest witness having  $\delta = 3$ ,  $\Delta = 5$  and 14 vertices. It is vacuous at  $\delta = 1$ , which is exactly where Hall's counterexample lives. The route runs through a bound of independent interest: every forest with matching number  $\nu$  satisfies  $m_{\nu-1}/m_\nu \leq \binom{\nu+1}{2}$ , with equality only for  $P_{2\nu}$  (Theorem 20), proved by Cauchy interlacing and a theorem of Gutman.

We also show that the obvious attack on the two repairs cannot work. A separator of size  $k$  met by  $p$  identical branches forces  $A^{p-k} \mid \mu_G$ , where  $A$  is the branch's matching polynomial (Lemma 7), so it produces roots of  $\mu_G$  at will; but when the branch is a tree that same configuration is a  $\theta$ -Aomoto subset in the sense of Banks, Garza-Vargas and Mukherjee, so the root it produces is an *eigenvalue* of the cover (Proposition 12). The condition  $p > k$  that makes the divisor nontrivial is literally the Aomoto inequality  $|\partial S| < \text{cc}(G[S])$ . The mechanism manufactures points of  $\text{spec}(T)$ , not counterexamples.

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This explains the one feature of Hall’s family that had resisted explanation, the pendant leaves. They create the root  $\sqrt{5}$  and simultaneously double the boundary of the relevant star, carrying it away from the Aomoto inequality; removing them destroys both. A counterexample must give its branches a private boundary, which is a condition on attachment rather than on degrees, and that is where a repair should be sought.

Everything asserted is machine-checked in Lean 4/Mathlib unless labelled otherwise. Supplementary material is in [18]; the biregular case is [17].

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**Code and data availability.** All computations, the Lean 4/Mathlib sources, and the scripts backing every numerical statement are in the `rama-notes` repository. Each numerical claim names the script that produced it; every named script is executed and its output compared against a recorded baseline by `code/cite_check.py`, and every Lean declaration cited is checked for its axiom base by `code/audit_papers.py`.

**Status conventions.** Claims here are of three kinds, and the text says which each time rather than leaving it to the reader. A result described as *formalized*, or cited by a Lean declaration name, is machine-checked in Lean 4/Mathlib: the sources are in `RamaLean/` of the accompanying repository, carry no `sorry`, and rest only on the standard axioms `propext`, `Classical.choice` and `Quot.sound`. A claim described as *measured*, *verified to* a stated tolerance, or *checked numerically* is computational evidence and is not a proof; the script producing it is named, and every named script is run and its output compared against a recorded baseline (`code/cite_check.py`). A claim called a *conjecture* is unproved, and is called that even where the evidence is strong.

## 1 Introduction and statement

The expected characteristic polynomial of a random lift is the object at the heart of the interlacing-families method of Marcus–Spielman–Srivastava [11], who used the  $r = 2$  (signing) case to prove that bipartite Ramanujan graphs exist in every degree; Hall–Puder–Sawin [2] extended the framework to arbitrary coverings. We evaluate it in closed form for the cycle.

A *permutation  $r$ -lift* of a graph  $G$  replaces each vertex by  $r$  copies and each edge  $\{u, v\}$  by the perfect matching  $u_i \sim v_{\sigma(i)}$  of a permutation  $\sigma \in S_r$  chosen independently and uniformly per edge. For the cycle  $C_n$ , gauge-fixing a spanning path reduces the lift to a single permutation  $\sigma \in S_r$  on the closing edge, and the lift decomposes into disjoint cycles: each  $\ell$ -cycle of  $\sigma$  contributes a cycle  $C_{n\ell}$  (fixed points contribute copies of  $C_n$ ). Since  $\chi_{C_m}(x) = \det(xI - A_{C_m}) = 2T_m(x/2) - 2$ , the expected characteristic polynomial is the expected cycle weight

$$\Phi_{n,r}(x) = \mathbb{E}_{\sigma \in S_r} \prod_{\ell \in \text{cycles}(\sigma)} (2T_{n\ell}(x/2) - 2). \quad (1)$$

**Theorem 1.** For all  $n \geq 1$ ,  $r \geq 1$ , with  $Y = T_n(x/2)$ ,

$$\Phi_{n,r}(x) = U_r(Y) - 2U_{r-1}(Y) + U_{r-2}(Y) = \underbrace{(2T_n(x/2) - 2)}_{\chi_{C_n}(x)} \cdot \underbrace{U_{r-1}(T_n(x/2))}_{\Psi_r(x)}.$$

The factored form identifies the quotient  $\Psi_r$ , by the theorem of Hall–Puder–Sawin [2] equating the expected new-eigenvalue characteristic polynomial with the  $d$ -matching polynomial ( $d = r - 1$ ), as

$$\mu_{d,C_n}(x) = U_d(T_n(x/2)),$$

which is equivalent (through  $\mathcal{P}_d(2t) = U_d(t)$ ,  $\mathcal{C}_n(2t) = 2T_n(t)$  and the composition identity  $U_{dn+n-1} = U_d(T_n)U_{n-1}$ ) to *Hall’s conjecture*  $\mathcal{C}_{n,d} = \mathcal{P}_d(\mathcal{C}_n)$ , proved combinatorially in [3]. That preprint has remained unrefereed since 2018 and carries no journal reference or DOI as of August 2026. The reason is not mathematical. Chris Hall, who posed the conjecture, has told us (personal communication, August 2026) that Andrew Herring was his first doctoral student; Hall gave him the conjecture, Herring presented it as a problem at a workshop, and the group of [3] solved it there. They submitted the result, met a couple of rejections, and abandoned the attempt for want of motivation rather than for any defect. Hall regards the proof as correct, and so do we. We record the episode because a reader weighing what is established about this identity should know both that the combinatorial proof stands and that the present derivation is, as far as we can tell, the first refereed or machine-checked one. Our route is different and elementary; we make no use of matching-polynomial combinatorics.

## What this paper does

The closed form is the starting point rather than the subject. Read as a statement about where the roots of  $\mu_{d,G}$  sit, it says that for the cycle they land exactly in  $\text{spec}(T)$ , and not merely in the interval  $[-\rho, \rho]$  that Hall–Puder–Sawin guarantee. The two differ whenever  $T$  has a spectral gap, and whether the stronger statement holds in general is the question here.

§3 proves it for every graph of first Betti number one and every  $d$ . §4 states it as a conjecture for all graphs. §5 is the counterexample, due to Chris Hall. §5.3 isolates the algebra behind it, and §5.4 shows that this algebra cannot be turned against the two natural repairs, because the configuration it needs is exactly an Aomoto subset. §5.5 and §5.7 contain the main theorem, the lower bound on the point spectrum in terms of the degrees, and §5.8 shows it is attained for every admissible pair of parameters. Supplementary material is in [18]; the biregular case, which survives the refutation, is [17].

## 2 Proof

Write  $f(\ell) = 2T_\ell(Y) - 2$  with  $Y = T_n(x/2)$ ; by the composition identity  $T_{n\ell} = T_\ell \circ T_n$  this is the per-cycle factor in (1).

*Step 1 (exponential formula).* For any commutative ring  $R$  and  $f : \mathbb{N} \rightarrow R$ , the  $S_r$ -average  $\Phi_r = \mathbb{E}_\sigma \prod_{\ell \in \text{cycles}(\sigma)} f(\ell)$  satisfies

$$r \Phi_r = \sum_{k=1}^r f(k) \Phi_{r-k}, \quad \Phi_0 = 1, \tag{2}$$

the coefficient form of  $\sum_r \Phi_r z^r = \exp(\sum_\ell f(\ell) z^\ell / \ell)$ . (Proof: group  $\sigma$  by cycle type  $\lambda$  with Cauchy's count  $r!/z_\lambda$ , mark a part, and re-index; see the Lean file `Paper2ExpFormula.lean` for the fully detailed version.)

*Step 2 (Chebyshev generating function).* With  $f(\ell) = 2T_\ell(Y) - 2$ ,

$$\sum_{\ell \geq 1} f(\ell) \frac{z^\ell}{\ell} = -\log(1 - 2Yz + z^2) + 2\log(1 - z) \implies \sum_{r \geq 0} \Phi_r z^r = \frac{(1 - z)^2}{1 - 2Yz + z^2}.$$

Extracting coefficients against  $\sum U_r(Y) z^r = (1 - 2Yz + z^2)^{-1}$  gives  $\Phi_r = U_r - 2U_{r-1} + U_{r-2}$ , and the three-term recurrence  $U_r + U_{r-2} = 2YU_{r-1}$  collapses this to  $(2Y - 2)U_{r-1}(Y)$ .  $\square$

In the formalization the analytic Step 2 is replaced by an equivalent finite argument: the closed form is shown to satisfy (2) directly (a second-difference induction), and (2) with  $\Phi_0 = 1$  determines  $\Phi$  over a characteristic-zero domain.

### 3 Graphs of first Betti number one

The proof above uses cycles in exactly one place: gauge-fixing a spanning path leaves a *single* permutation, because  $C_n$  has  $|E| - |V| + 1 = 1$ . The argument therefore applies verbatim to any connected  $G$  with  $b_1(G) = |E| - |V| + 1 = 1$ , a cycle with trees attached, and we record what it gives.

By [2] the expected characteristic polynomial of a random  $r$ -lift of *any* graph factors as  $\chi_G \cdot \mu_{r-1,G}$ ; the content below is therefore not that factorization but a closed form for the factor  $\mu_{d,G}$  itself. Question 6.5 of [2] asks which parts of the theory of the matching polynomial  $\mu_{1,G}$  generalize to  $\mu_{d,G}$ ; the following answers it for  $b_1(G) = 1$ .

Let  $C$  be the unique cycle of  $G$  and let  $G - V(C)$  be the forest obtained by deleting its vertices. Write  $\mu_H$  for the ordinary matching polynomial of a graph  $H$ , with  $\mu_\emptyset = 1$ . Define

$$V_0 = 1, \quad V_1 = \mu_G, \quad V_d = \mu_G V_{d-1} - \mu_{G-V(C)}^2 V_{d-2}. \quad (3)$$

**Theorem 2.** *For every connected  $G$  with  $b_1(G) = 1$  and every  $d \geq 0$ ,  $\mu_{d,G} = V_d$ .*

Theorem 2 is proved below, through the twisted characteristic polynomial; it is not formalized.

For  $G = C_n$  the deleted forest is empty, so  $\mu_{G-V(C)} = 1$  and (3) is the recurrence for  $U_d(\mu_{C_n}/2) = U_d(T_n(x/2))$ : Theorem 21 is the case  $\mu_{G-V(C)} = 1$ . In general the attached trees enter only through  $\mu_{G-V(C)}$ ; for the triangle with one pendant edge,  $\mu_G = x^4 - 4x^2 + 1$  and  $\mu_{G-V(C)} = x$ .

The proof goes through the twisted characteristic polynomial. Let  $e = uv$  be the unique non-tree edge, let  $A_G(z)$  be the adjacency matrix of  $G$  with the  $(u, v)$  entry replaced by  $z$  and the  $(v, u)$  entry by  $z^{-1}$ , and set  $F(x, z) = \det(xI - A_G(z))$ .

**Lemma 3.**  $F(x, z) = \mu_G(x) - \mu_{G-V(C)}(x)(z + z^{-1})$ .

*Proof.* By the Sachs coefficient theorem  $\det(xI - A)$  expands over elementary subgraphs, that is disjoint unions of edges and cycles. Since a permutation uses each of the two twisted entries at most once and  $A_G(z)^\top = A_G(z^{-1})$ , the determinant is of the form  $P(x) + Q(x)(z + z^{-1})$ . Replacing  $z$  by  $\pm 1$  and averaging kills exactly the elementary subgraphs whose cycle traverses

$e$ ; as  $G$  has the single cycle  $C$ , those are the subgraphs containing  $C$ , so  $P = \mu_G$ , and the surviving terms contribute  $Q = -\mu_{G-V(C)}$ , a cycle component carrying the Sachs weight  $-2$ .  $\square$

*Proof of Theorem 2.* Gauge-fix a spanning tree: the  $r$ -lift is determined by one permutation  $\sigma \in S_r$  on  $e$ , and its components are indexed by the cycles of  $\sigma$ , the component of an  $\ell$ -cycle being the  $\ell$ -fold cyclic cover  $G_\ell$ . Decomposing the  $\mathbb{Z}/\ell$  action into characters gives the classical factorization  $\chi_{G_\ell}(x) = \prod_{j=0}^{\ell-1} F(x, \omega^j)$  with  $\omega = e^{2\pi i/\ell}$  [9, 10]. With  $P, Q$  as in Lemma 3 and  $y = -P/(2Q)$ ,

$$\chi_{G_\ell} = \prod_{j=0}^{\ell-1} \left( P + 2Q \cos \frac{2\pi j}{\ell} \right) = (-Q)^\ell (2T_\ell(y) - 2),$$

so by the exponential formula (2), with  $w = -Qz$ ,

$$\sum_{r \geq 0} \Phi_{G,r} z^r = \exp \left( \sum_{\ell \geq 1} (2T_\ell(y) - 2) \frac{w^\ell}{\ell} \right) = \frac{(1-w)^2}{1-2yw+w^2},$$

and extracting coefficients as in Section 2 gives  $\Phi_{G,r} = \chi_G \cdot (-Q)^{r-1} U_{r-1}(y)$ . Comparing with  $\Phi_{G,r} = \chi_G \cdot \mu_{r-1,G}$  [2] yields  $\mu_{d,G} = (-Q)^d U_d(y)$ , which satisfies (3).  $\square$

Since  $U_d(y) = 2^d \prod_{k=1}^d (y - \cos \frac{k\pi}{d+1})$ , the theorem can be restated as a product,  $\mu_{d,G}(x) = \prod_{k=1}^d F(x, e^{ik\pi/(d+1)})$ , whence:

This is the real-rootedness theorem of [2] for this family, with the roots identified. The localization itself, that a vanishing  $V_d$  forces  $|\mu_G| \leq 2|\mu_{G-V(C)}|$ , so that a root can never sit in a spectral gap, is machine-checked in Lean 4 as `cV_root_mem_band`, resting on `cU_ne_zero_of_one_le_abs`: the Chebyshev polynomial  $U_d$  has no root of absolute value at least one.

The roots can be located much more precisely than that. Since  $b_1(G) = 1$  the group  $\pi_1(G)$  is  $\mathbb{Z}$ , so the universal cover  $T$  of  $G$  is the  $\mathbb{Z}$ -cover, and  $T$  is a periodic decorated chain whose adjacency operator decomposes by Floquet–Bloch into the fibres  $A_G(e^{i\theta})$ . Hence

$$\text{spec}(T) = \bigcup_{\theta} \text{spec } A_G(e^{i\theta}) = \left\{ x : \exists s \in [-1, 1], \mu_G(x) = 2s \mu_{G-V(C)}(x) \right\},$$

a finite union of bands, together with a flat band at each common zero of  $\mu_G$  and  $\mu_{G-V(C)}$ , where  $F(x, \cdot)$  vanishes identically.

**Corollary 4.** *For  $b_1(G) = 1$  the roots of  $\mu_{d,G}$  are exactly the solutions of  $\mu_G(x) = 2 \cos(\frac{k\pi}{d+1}) \mu_{G-V(C)}(x)$  for  $1 \leq k \leq d$ . In particular every root of every  $\mu_{d,G}$  lies in  $\text{spec}(T)$ , and as  $d \rightarrow \infty$  the roots equidistribute on  $\text{spec}(T)$  with respect to its density of states.*

*Proof.* Immediate from Theorem 2 in the form  $\mu_{d,G} = \prod_k F(x, e^{ik\pi/(d+1)})$  and  $F(x, e^{i\theta}) = \mu_G - 2 \cos \theta \mu_{G-V(C)}$ . The equidistribution is the arcsine law for  $\cos(k\pi/(d+1))$  transported through the fibre map.  $\square$

Hall–Puder–Sawin place the roots of  $\mu_{d,G}$  in the Ramanujan *interval*  $[-\rho, \rho]$ ,  $\rho$  being the spectral radius of  $T$ . Corollary 4 places them in  $\text{spec}(T)$ , which is a union of bands and is a proper subset of that interval whenever  $T$  has a spectral gap: no root of any  $\mu_{d,G}$  can ever enter a gap. The two statements coincide exactly when the band structure is trivial, and that is precisely the case of a cycle, where  $\mu_{G-V(C)} = 1$  and  $\text{spec}(T) = [-2, 2]$  is the whole interval. Attaching even a single pendant edge opens gaps, and they are not small (computed band structures, proportion of  $[-\rho, \rho]$  that is a gap):

| $G$                  | $\rho$ | gap proportion |
|----------------------|--------|----------------|
| $C_n$                | 2      | 0%             |
| triangle + pendant   | 2.170  | 36.5%          |
| $C_4$ + pendant      | 2.136  | 45.9%          |
| triangle + $P_2$     | 2.214  | 51.3%          |
| $C_4$ + $P_3$        | 2.189  | 54.1%          |
| triangle + $K_{1,3}$ | 2.514  | 65.9%          |

So the cycle is not merely the easiest case of Theorem 2: it is the degenerate one, the unique member of the family whose spectrum has no gaps and for which the Ramanujan interval is attained. Any cycle with a tree attached exhibits a band structure that no cycle can display.

**The boundary is sharp.** For  $b_1(G) = b$  the lift is determined by  $b$  permutations, that is by an action of the free group  $F_b$  on  $[r]$ , whose orbits are the components of the lift. The exponential formula still applies, in the form  $N_r = \sum_{\ell} \binom{r-1}{\ell-1} C_{\ell} N_{r-\ell}$  with  $N_r = (r!)^b \Phi_{G,r}$  and  $C_{\ell}$  the sum of  $\chi$  over the transitive  $b$ -tuples on  $[\ell]$ ; we have verified this for  $b \leq 3$ . What fails is the atom. For  $b = 1$  every transitive tuple gives the *same* cover  $G_{\ell}$ , so  $C_{\ell} = (\ell - 1)! \chi_{G_{\ell}}$ : one graph per  $\ell$ , with  $\chi_{G_{\ell}}$  Chebyshev in  $\ell$ . For  $b \geq 2$  the transitive tuples number  $(\ell - 1)!$  times the count of index- $\ell$  subgroups of  $F_b$  and spread over many non-isomorphic covers with distinct spectra, so  $C_{\ell}$  carries no such structure. Accordingly Theorem 2 is not merely unproved for  $b \geq 2$ : it is false there already at  $d = 1$ , as we checked for the theta graph, the bowtie,  $K_4$ ,  $K_{2,3}$  and two triangles joined by an edge.

Theorem 2 was verified by exact integer computation against the definition of  $\mu_{d,G}$  as an average of matching polynomials over all  $d$ -covers, and against brute-forced lift averages, on thirteen named unicyclic graphs for  $r \leq 6$  and on a random sample of thirty-four for  $r \leq 5$ ; Lemma 3 was checked on eleven.

## 4 A conjecture for all graphs

Corollary 4 is proved by Floquet theory, which is available only because  $\pi_1(G) = \mathbb{Z}$ . Its *statement*, however, makes sense for every graph, and we know one case of it: at  $b_1(G) = 1$  it is Corollary 4, for every  $d$ . We therefore propose:

**Conjecture 5.** *For every finite graph  $G$  and every  $d \geq 1$ , all roots of  $\mu_{d,G}$  lie in  $\text{spec}(T)$ , the spectrum of the universal cover of  $G$ .*

**Theorem 6** (Hall [12]). *Conjecture 5 is false, already at  $d = 1$  and already for simple connected loopless bipartite graphs.*

The counterexample and its certificate are entirely due to Chris Hall. We record it here because it settles the central question of this paper, and because everything else in the note has to be read in its light. §5 states his construction, our independent verification of it, and what we can add.

The biregular case of Conjecture 5 at  $d = 1$  is not new. It is Problem 1 of Song, Fan and Miao [5], posed in 2023: they ask whether the matching polynomial of the incidence graph  $B_H$  of a  $(d, r)$ -regular hypergraph has no root  $\mu$  with  $0 < |\mu| < |\sqrt{d-1} - \sqrt{r-1}|$ , and every  $(d, r)$ -biregular bipartite graph is such an incidence graph. Their Theorem 1.8 attaches a consequence: an affirmative answer would give every  $(d, r)$ -regular hypergraph a left-sided Ramanujan 2-covering. The only bound in that direction we know of is their Lemma 5.5, which goes the other way,  $0 < \mu_\tau(B_H) \leq \max\{\sqrt{d}, \sqrt{r}\}$ . We are not aware of any proof, partial result or counterexample. Conjecture 5 extends their problem in two directions, to arbitrary graphs and to all  $d$ .

Question 6.3 of [2] asks the corresponding question for graph eigenvalues, and leaves it open: it distinguishes graphs whose nontrivial spectrum lies in the interval  $[-\rho, \rho]$  from those whose nontrivial spectrum lies in  $\text{spec}(T)$ , observes that the two notions separate exactly when  $T$  has a gap, and gives the biregular tree as the example. Conjecture 5 is the matching-polynomial form of that distinction.

The conjecture strengthens the root-localization theorem of [2], which places the roots in the interval  $[-\rho, \rho]$  with  $\rho$  the spectral radius of  $T$ ; the conjecture places them in  $\text{spec}(T)$  itself. The two agree whenever  $T$  has no spectral gap, in particular for regular  $G$ , where  $T$  is a regular tree and  $\text{spec}(T) = [-2\sqrt{q}, 2\sqrt{q}]$  is an interval. They differ exactly when  $T$  has a gap, and the smallest examples are biregular: for  $K_{2,3}$  the universal cover is the  $(3, 2)$ -biregular tree, with spectrum  $\{0\} \cup \pm[\sqrt{2}-1, \sqrt{2}+1]$  and hence a gap  $(0, \sqrt{2}-1)$  inside the Ramanujan interval. We have tested the conjecture on the biregular family, where the universal cover is an  $(a, b)$ -biregular tree with spectrum  $\{0\} \cup \pm[|s-t|, s+t]$ ,  $s = \sqrt{a-1}$ ,  $t = \sqrt{b-1}$ , so the gap is  $(0, |s-t|)$ . In each case  $\mu_{d,G}$  was computed by averaging matching polynomials over all  $d$ -covers. No root ever lies in the gap; we also record the smallest nonzero  $|\text{root}|$  as a multiple of the gap edge, which measures how close the bound comes to being attained.

| $G$                  | $b_1$ | type   | gap edge | $d$        | $\min  \text{root} /\text{gap edge}$ |
|----------------------|-------|--------|----------|------------|--------------------------------------|
| $K_{2,3}$            | 2     | (3, 2) | 0.4142   | 1, 2, 3, 4 | 2.72, 2.19, 1.94, 1.78               |
| $K_{2,4}$            | 3     | (4, 2) | 0.7321   | 1, 2, 3    | 1.93, 1.65, 1.52                     |
| $K_{2,5}$            | 4     | (5, 2) | 1.0000   | 1, 2, 3    | 1.66, 1.46, 1.37                     |
| $K_{3,4}$            | 6     | (4, 3) | 0.3178   | 1, 2       | 3.04, 2.31                           |
| $K_{3,5}$            | 8     | (5, 3) | 0.5858   | 1, 2       | 2.10, 1.73                           |
| subdivision of $K_4$ | 3     | (3, 2) | 0.4142   | 1, 2       | 1.97, 1.67                           |

All of these have  $b_1 \geq 2$ , so none is covered by Corollary 4. It is worth recording that even  $d = 1$  is not formal. There  $\mu_{1,G}$  is the ordinary matching polynomial, whose roots are eigenvalues of the path tree of  $G$  (Godsil), and the path tree embeds in  $T$ ; but eigenvalues of a finite subtree of  $T$  need not lie in  $\text{spec}(T)$  when  $T$  has a gap. For the  $(3, 2)$ -biregular tree, whose gap is  $(0, \sqrt{2}-1)$ , the path  $P_8$  is a subtree with eigenvalue  $2 \cos(4\pi/9) = 0.3473$  inside that gap. So the  $d = 1$  case does not follow from the path-tree argument. It is nevertheless a theorem for an infinite family.

## 5 Hall's counterexample

Everything in this section is due to Chris Hall, who constructed the graph, found the certificate, and wrote the verification note [12]. Our contribution is the independent check, the formalization of the algebra. The two-parameter family containing his graph is in [18].

### 5.1 The graph and its matching polynomial

Let  $G$  consist of a central vertex  $c$ ; for  $1 \leq i \leq 5$  a copy of  $K_{2,5}$  with degree-five vertices  $v_i, w_i$  and middle vertices  $u_{i,1}, \dots, u_{i,5}$ ; a pendant leaf  $\ell_i$  at each  $w_i$ ; and the edges  $cv_i$ . It is simple, connected, loopless and bipartite, with 41 vertices, 60 edges and  $b_1(G) = 20$ .

Writing  $H$  for one branch after deleting  $c$ , the vertex-deletion recurrence gives  $\mu_{H-v} = x^5(x^2 - 6)$  and  $\mu_H = x^4(x^4 - 11x^2 + 25)$ , and then

$$\mu_G(x) = \mu_H(x)^4 (x\mu_H(x) - 5\mu_{H-v}(x)) = x^{21}(x^4 - 11x^2 + 25)^4(x^2 - 5)(x^2 - 11), \quad (4)$$

so  $\sqrt{5}$  is a simple root. Identity (4), the simplicity of the root, and the value of the cofactor are machine-checked in `RamaLean/HallCounterexample.lean` as an identity in  $\mathbb{Z}[X]$  (`muG_eq`, `muG_root_sqrt5`, `muG_root_simple`).

### 5.2 The spectral exclusion

Angel, Friedman and Hoory [13] characterise invertibility of  $A_T - \lambda I$  by a finite nonzero *ratio system*: an assignment  $r_e$  to directed edges with  $\lambda = 1/r_e + \sum_{e \rightarrow f} r_f$ , whose decay matrix  $\mathcal{R}_{e,f} = |r_f|^2$  on non-backtracking pairs has spectral radius below one. Hall exhibits one at  $\lambda = \sqrt{5}$  with ratios in  $\mathbb{Q}(\sqrt{5}, \sqrt{41})$ ; its 120-state decay matrix has ten transient states at the leaves, and the recurrent block of 110 reduces by symmetry to a  $6 \times 6$  orbit quotient  $K$  with  $x = (20, 121, 47, 33, 28, 16)^\top$  satisfying  $Kx < x$  componentwise, so  $\rho(K) < 1$  by Perron–Frobenius. Hence  $\sqrt{5} \notin \text{spec}(A_T)$ , which with (4) proves Theorem 6. We rebuilt every object from the graph rather than copying the displayed matrices (`code/hall_certificate.py`) and machine-checked the algebra (`RamaLean/HallCounterexample.lean`).

### 5.3 The Divisibility Lemma

The counterexamples are all instances of one algebraic fact, which we state once.

**Lemma 7** (Divisibility). *Let  $G$  be a finite graph,  $S \subseteq V(G)$  with  $|S| = k$ , and suppose  $G - S$  has  $p$  components, each inducing a copy of the same graph  $B$ . Write  $A = \mu_B$ . Then*

$$A^{p-k} \mid \mu_G.$$

*Proof.* Expand  $\mu_G = \sum_M (-1)^{|M|} x^{n-2|M|}$  over matchings  $M$  of  $G$  and group the terms by

$$J(M) = \{i : \text{some edge of } M \text{ joins } S \text{ to the } i\text{-th component}\}.$$

A matching uses each vertex of  $S$  at most once, so distinct components in  $J(M)$  are entered through distinct vertices of  $S$ ; choosing one such vertex per component is an injection  $J(M) \hookrightarrow S$ , whence  $|J(M)| \leq k$ . For  $i \notin J(M)$  the restriction of  $M$  to the  $i$ -th component

is an arbitrary matching of  $B$ , and summing over those choices independently contributes a factor  $A^{p-|J(M)|}$ . Each term of the resulting sum is therefore divisible by  $A^{p-k}$ , and hence so is  $\mu_G$ .  $\square$

At  $k = 2$  this is the divisor  $A^{p-2}$  of the two-cut identity. The exponent is positive exactly when  $p > k$ , which is why every construction below needs at least  $k + 1$  branches. The counting step and the algebraic step are machine-checked separately (RamaLean/DivisibilityLemma.lean: `card_branches_le_sep`, `dvd_sum_of_pow_terms`, `matching_expansion_dvd`); the matching expansion itself is not, Mathlib carrying no matching polynomial.

## 5.4 Why the branch mechanism cannot produce a counterexample

The natural repairs are local hypotheses on  $G$ . Minimum degree two fails and bounded maximum degree fails, so one is led to:

**Conjecture 8 (D3).** *Every finite graph of minimum degree at least three satisfies  $\text{Zeros}(\mu_G) \subseteq \text{spec}(T_G)$ .*

**Conjecture 9 (C2).** *Every finite 3-connected graph satisfies  $\text{Zeros}(\mu_G) \subseteq \text{spec}(T_G)$ .*

Both are open. We record here why the construction of §5.3, which is the obvious way to attack them, cannot settle either: the configuration that makes Lemma 7 produce a root of  $\mu_G$  is the same configuration that puts that root *inside*  $\text{spec}(T_G)$ .

Banks, Garza-Vargas and Mukherjee [14] characterise the eigenvalues of the universal cover:  $\theta$  is an eigenvalue of  $T_G$  if and only if  $G$  has a  $\theta$ -Aomoto subset, that is a set  $S \subseteq V(G)$  with  $G[S]$  a forest,  $\theta$  an eigenvalue of every component of  $G[S]$ , and

$$|\partial_G S| < \text{cc}(G[S]).$$

Li, Magee, Sabri and Thomas [15] prove the corresponding statement for the maximal abelian cover, in the form that  $\theta$  is an eigenvalue of  $G^{\text{ab}}$  if and only if  $\theta$  is a zero of  $\mu_{G \setminus \Gamma}$  for every 2-regular subgraph  $\Gamma$ , and Spier [16] proves that  $T_G$  and  $G^{\text{ab}}$  have the same eigenvalues.

Combining the two gives an exact and purely combinatorial certificate, which is worth stating because it replaces every numerical test used in [18]. The empty subgraph is 2-regular, so taking  $\Gamma = \emptyset$  recovers the containment of the point spectrum in the zeros of  $\mu_G$ ; the content is the converse direction.

*Remark 10.* A root  $\theta$  of  $\mu_G$  fails to be an eigenvalue of  $T_G$  if and only if there is a 2-regular subgraph  $\Gamma \subseteq G$ , that is a disjoint union of cycles, with  $\mu_{G \setminus \Gamma}(\theta) \neq 0$ . A counterexample to Conjecture 5 is therefore a graph carrying a root  $\theta$  of  $\mu_G$  that lies outside the bands of  $\text{spec}(T_G)$  together with one such witness  $\Gamma$ , and both halves are finite exact checks on polynomials.

This is the check that should have been run in [18]. The density-of-states and ratio-system tests used there decide eigenvalue membership numerically, and it was exactly that step which produced the withdrawn claims about Conjectures 8 and 9: in each case the root was an eigenvalue after all, carried back into the spectrum by an Aomoto subset that the numerics did not see. Running it settles both sides.

For the 14-vertex construction of [18], two hubs with three copies of  $K_{1,3}$  and every leaf joined to both hubs, the graph has 138 cycles and 193 two-regular subgraphs, and  $\mu_{G \setminus \Gamma}(\sqrt{3}) = 0$  for *every one of them*. So  $\sqrt{3}$  is an eigenvalue of  $T_G$ , the retraction is confirmed without any numerical input, and the construction was not a near miss: there is no witness to be found.

For Hall's graph the certificate is short. Write  $B$  for one block, a copy of  $K_{2,5}$  with a pendant leaf on one of the two vertices of degree 5, joined to the centre at the other. Every cycle of the graph lies inside a block and is a 4-cycle, since  $K_{2,5}$  is bipartite with one part of size two, and any two of them meet in that part; so a two-regular subgraph takes at most one cycle per block. Deleting one such cycle isolates the rest of its block and severs it from the centre, and the cut-vertex recursion at the centre gives  $\mu_{H_s} = x\mu_B^s - s\mu_{B-v}\mu_B^{s-1}$  for the centre joined to  $s$  intact blocks. Here  $\mu_B(\sqrt{5}) = -125$  and  $\mu_{B-v}(\sqrt{5}) = -25\sqrt{5}$ , so  $\mu_G(\sqrt{5}) = 0$  while a single 4-cycle already gives  $\mu_{G \setminus \Gamma}(\sqrt{5}) = 1220703125\sqrt{5} \neq 0$ .

**Proposition 11.** *In Hall's 41-vertex graph,  $\sqrt{5}$  is a root of  $\mu_G$  and is not an eigenvalue of  $T_G$ , and a single 4-cycle inside any one block witnesses this.*

The point is not that the conclusion changes, it does not, but that it is now certified by integer arithmetic in a quadratic field rather than by an estimate of the width of a spectral gap.

**Proposition 12** (the branch obstruction). *Let  $\theta \in \mathbb{R}$  and let  $S \subseteq V(G)$  with  $|S| = k$ . If at least  $k + 1$  of the components of  $G - S$  induce forests having  $\theta$  as an eigenvalue, then  $\theta$  is an eigenvalue of  $T_G$ , and in particular  $\theta \in \text{spec}(T_G)$ .*

*Proof.* Let  $U$  be the union of  $j \geq k + 1$  such components. Then  $G[U]$  is a disjoint union of forests, hence a forest, with  $\text{cc}(G[U]) = j$  and  $\theta$  an eigenvalue of every component. Every vertex outside  $U$  adjacent to  $U$  lies in  $S$ , so  $|\partial_G U| \leq k < j$ . Thus  $U$  is a  $\theta$ -Aomoto subset and [14] applies.  $\square$

The components need not be isomorphic to one another, and need not be all of  $G - S$ ; only their number relative to  $|S|$  matters. Taking them to be  $p$  copies of a single tree  $B$  with  $\theta$  a root of  $\mu_B$  recovers the case that concerns us:

The hypotheses of Proposition 12 are then exactly those of Lemma 7 together with  $p > k$ , which is what the lemma needs for the exponent  $p - k$  to be positive. So whenever the branch mechanism forces a root of  $\mu_G$ , it forces that root into  $\text{spec}(T_G)$ . The two conditions are the same condition:

$$p > k \quad \text{is} \quad |\partial_G S| < \text{cc}(G[S]).$$

A shared separator keeps the boundary at  $k$  while the number of components grows with  $p$ , and the Aomoto inequality is then automatic.

This is why Hall's construction is subtler than it looks, and it explains the role of the pendant leaves, which [18] records as essential without a mechanism. His branches are not trees, so the obstruction does not apply to them directly; the natural forest inside a branch is the star  $K_{1,5}$  at  $w_i$ , whose boundary is the pair  $\{v_i, \ell_i\}$ . With five branches that gives  $\text{cc} = 5$  and  $|\partial| = 10$ , comfortably on the wrong side of the Aomoto inequality. Deleting the pendant leaves brings the boundary down to 5, still not below 5, and it also destroys the root: the leafless graph has  $\mu = x^{16}(x^4 - 15x^2 + 45)(x^4 - 10x^2 + 20)^4$ , with no factor  $x^2 - 5$ . The

leaves therefore do two jobs at once, creating the root and doubling the boundary of the star, and both are needed (`code/aomoto_obstruction.py`). The general form is checked there on non-isomorphic branches as well, a hub joined to both ends of a  $P_3$  and of a  $P_7$ , where  $\sqrt{2}$  is an eigenvalue of each path and the two components already outnumber the single boundary vertex.

The moral for a repair is that it must forbid the *private* boundary structure that Hall uses. Every counterexample must avoid having a  $\theta$ -Aomoto subset at its offending root, which is primarily a condition on how branches attach. The ambient degrees are not irrelevant, however: they enter through a counting bound, which we record next because it cuts the search space in half.

## 5.5 A degree bound on the point spectrum

**Proposition 13** (degree spread). *Let  $G$  be a graph with minimum degree  $\delta \geq 2$  and maximum degree  $\Delta$ . If  $T_G$  has a nonzero eigenvalue, then*

$$\Delta > 2(\delta - 1), \quad \text{equivalently} \quad \Delta \geq 2\delta - 1.$$

*Proof.* By [14] there is a  $\theta$ -Aomoto subset  $S$  with  $\theta \neq 0$ . Put  $s = |S|$ ,  $c = \text{cc}(G[S])$ ,  $b = |\partial S|$ , and let  $e$  be the number of edges of  $G$  with exactly one end in  $S$ . Since  $G[S]$  is a forest with  $c$  components it has  $s - c$  edges, so counting the degrees of the vertices of  $S$  gives  $e = \sum_{v \in S} \deg v - 2(s - c) \geq (\delta - 2)s + 2c$ . Every such edge has its other end in  $\partial S$ , and a boundary vertex meets at most  $\Delta$  edges in total, so  $e \leq \Delta b$ . As  $\theta \neq 0$ , no component of  $G[S]$  is a single vertex, whence  $s \geq 2c$ ; and the Aomoto inequality is  $b \leq c - 1$ . Combining,

$$2c(\delta - 1) = (\delta - 2)(2c) + 2c \leq (\delta - 2)s + 2c \leq e \leq \Delta b \leq \Delta(c - 1) < \Delta c,$$

and dividing by  $c > 0$  gives  $2(\delta - 1) < \Delta$ . □

Taking  $\Delta = \delta$  recovers the classical fact that a regular tree has no eigenvalues. The case that matters here is the contrapositive.

**Corollary 14.** *If  $\delta(G) \geq 3$  and  $\Delta(G) \leq 2\delta(G) - 2$ , then  $T_G$  has no nonzero eigenvalue. In particular this holds for every graph with degrees in  $\{3, 4\}$ .*

The counting step and both consequences are machine-checked in `RamaLean/DegreeBound.lean` as `degree_bound`, `no_eigenvalue_of_small_spread` and `mindeg_three_maxdeg_four`; as in Proposition 12, the criterion of [14] is the analytic input and is cited rather than reproved, and the three combinatorial facts above enter the Lean statement as hypotheses.

Corollary 14 is what makes the bound useful rather than decorative. A counterexample to Conjecture 8 needs a root of  $\mu_G$  that is outside the bands of  $\text{spec}(T_G)$  and not an eigenvalue, and it was the second condition that defeated every attempt in [18]: each time, the root that appeared to escape was carried back into the spectrum by an Aomoto subset. In the range  $\Delta \leq 2\delta - 2$  that failure mode is impossible, so only the first condition remains. Every known counterexample to Conjecture 5, Hall's included, has  $\delta = 1$ , for which the bound is vacuous; this is consistent with the role his pendant leaves play above.

## 5.6 The origin: Hall's condition, after Banks–Garza–Vargas–Mukherjee

Proposition 13 needs  $\theta \neq 0$ , because the step  $s \geq 2c$  used that no component of  $G[S]$  is a single vertex. The excluded case has a complete and entirely elementary answer, which is *not* new: it is contained in [14], whose Observation 4.2 states that for zero potential the Aomoto set at  $\lambda = 0$  is independent, every Aomoto tree being an isolated vertex. Combined with their Theorems 3.1 and 3.3 this gives the following at once. We record it because the phrasing in terms of Hall's condition is convenient later, and we give the elementary proof because it is self-contained, but no claim of novelty is made for it.

**Proposition 15** ([14, Observation 4.2 with Theorems 3.1 and 3.3]). *For any graph  $G$ , the following are equivalent.*

1.  $0$  is an eigenvalue of the universal cover  $T_G$ .
2.  $G$  contains an independent set  $S$  with  $|N(S)| < |S|$ , that is,  $G$  fails Hall's condition.

*Proof.* By [14], (i) holds iff  $G$  has a 0-Aomoto subset.

If  $S$  is independent then every component of  $G[S]$  is a single vertex, whose only eigenvalue is 0, and  $\text{cc}(G[S]) = |S|$  while  $\partial_G S = N(S)$ . So the Aomoto condition  $|\partial_G S| < \text{cc}(G[S])$  reads  $|N(S)| < |S|$ , and (ii) implies (i).

Conversely let  $S$  be a 0-Aomoto subset, with  $G[S]$  a forest of components  $T_1, \dots, T_c$  and  $b := |\partial_G S| \leq c - 1$ . A forest has 0 as an eigenvalue iff it has no perfect matching, so each  $T_i$  has deficiency  $d_i := |T_i| - 2\nu_i \geq 1$ , where  $\nu_i$  is its matching number. Choose a maximum independent set  $S'_i \subseteq T_i$ ; as  $T_i$  is bipartite, König's theorem gives  $|S'_i| = |T_i| - \nu_i = \nu_i + d_i$ , and  $N_{T_i}(S'_i) \subseteq T_i \setminus S'_i$  has at most  $\nu_i$  elements. Put  $S' = \bigcup_i S'_i$ . Distinct components of an induced subgraph have no edges between them, so  $S'$  is independent in  $G$ , and every neighbour of  $S'$  lies in some  $T_i \setminus S'_i$  or in  $\partial_G S$ . Hence

$$|S'| = \sum_i (\nu_i + d_i) \geq \sum_i \nu_i + c, \quad |N(S')| \leq \sum_i \nu_i + b \leq \sum_i \nu_i + c - 1,$$

so  $|N(S')| < |S'|$  and (ii) holds.  $\square$

The argument below is a re-derivation by counting; the proof in [14] is shorter and stronger, deducing from their Lemma 4.2 that each Aomoto tree carries a nowhere-vanishing  $\lambda$ -eigenvector, and that a kernel vector of a zero-potential tree operator cannot be nonzero at the parent of a leaf, so the Aomoto trees at  $\lambda = 0$  are single vertices. The counting step is machine-checked in `RamaLean/HallZero.lean` as `hall_zero_counting` and `hall_violator_of_aomoto_zero`; as in Proposition 13, the criterion of [14] is the analytic input and is cited rather than reproved, and König's theorem and the deficiency of a forest without a perfect matching enter as hypotheses. The equivalence itself, and every step of the construction above, was checked by exhaustive enumeration of all 27474 connected graphs on at most six vertices, with 2765 positive instances and no discrepancy (`code/hall_zero.py`).

Two consequences are worth stating. First, an independent set violating Hall's condition admits no matching saturating it, so Theorem 15 gives at once that  $0 \in \text{spec}_p(T_G)$  implies  $G$  has no perfect matching, hence  $\mu_G(0) = 0$ . The converse fails:  $C_5$  has no perfect matching and no Hall violator, and indeed its cover is the infinite path, whose spectrum  $[-2, 2]$  contains 0 but not as an eigenvalue. Second, taken with Proposition 13, the point spectrum of a

universal cover is now bounded by combinatorics from both sides: at the origin it is exactly a Hall violation, and away from the origin it requires the degree spread  $\Delta > 2(\delta - 1)$ .

## 5.7 Small eigenvalues need large Aomoto trees

The step  $s \geq 2c$  in Proposition 13 is the weakest link in the chain, and it is weak by a wide margin. It uses only that a  $\theta$ -Aomoto tree with  $\theta \neq 0$  is not a single vertex. Lemma 4.2 of [14] says much more: each component  $T_i$  of  $G[S]$  carries a  $\theta$ -eigenvector that is nonzero at *every* vertex. Such a tree is precisely a  $\theta$ -critical tree in the sense of Ku and Wong, the notion under which Spier [16, Cor. 40] identifies the components of a refined  $\theta$ -Aomoto subset, and  $\kappa(\theta)$  below is the least order of one. Such trees cannot be small when  $\theta$  is small. The only tree of order 2 is  $K_2$ , whose eigenvalues are  $\pm 1$ ; the two trees of order 3 are  $P_3$ , with  $\pm\sqrt{2}$  and 0, and the eigenvector for 0 vanishes at the centre. So already  $0 < |\theta| < 1$  forces every component to have at least four vertices.

Write  $\kappa(\theta)$  for the least order of a tree possessing a nowhere-vanishing  $\theta$ -eigenvector. Then  $s \geq \kappa(\theta)c$ , and the same counting chain gives the following.

**Proposition 16** (refined degree spread). *Let  $\delta(G) \geq 2$  and let  $\theta \neq 0$  be an eigenvalue of  $T_G$ . Then*

$$\Delta > (\delta - 2)\kappa(\theta) + 2.$$

*Proof.* Verbatim the proof of Proposition 13 with  $s \geq \kappa c$  in place of  $s \geq 2c$ :  $c((\delta - 2)\kappa + 2) \leq (\delta - 2)s + 2c \leq e \leq \Delta b \leq \Delta(c - 1) < \Delta c$ .  $\square$

Proposition 13 is the case  $\kappa = 2$ . The content is that  $\kappa(\theta)$  grows as  $\theta \rightarrow 0$ , and the path family shows how fast.

**Proposition 17** (paths). *The eigenvector of  $P_n$  for  $\lambda_k = 2 \cos \frac{k\pi}{n+1}$  has  $j$ -th entry  $\sin \frac{jk\pi}{n+1}$ , so it is nowhere vanishing if and only if  $\gcd(k, n+1) = 1$ . In particular  $P_{2m}$  carries a nowhere-vanishing eigenvector for*

$$\lambda_m = 2 \cos \frac{m\pi}{2m+1} = 2 \sin \frac{\pi}{2(2m+1)},$$

since  $\gcd(m, 2m+1) = 1$ , and hence  $\kappa(\theta) \leq 2m$  for every  $|\theta| \geq 2 \sin \frac{\pi}{2(2m+1)}$ .

The values are 1, 0.6180, 0.4450, 0.3473, 0.2846, 0.2411 for  $m = 1, \dots, 6$ , and  $2 \sin \frac{\pi}{2(2m+1)} \sim \pi/(2m+1)$ , so this family gives  $\kappa(\theta) \lesssim \pi/|\theta|$ . An exhaustive computation over all trees of order at most 19 finds nothing better: the minimum of  $|\theta|$  over all nowhere-vanishing eigenpairs of all trees of order at most  $n$  equals  $2 \cos \frac{m\pi}{2m+1}$  for  $n \in \{2m, 2m+1\}$ ,  $m \leq 9$ , and is attained by  $P_{2m}$ . Odd orders contribute nothing new; the minimum over trees of order exactly  $2m+1$  is strictly larger than the minimum over trees of order exactly  $2m$ , and for odd order it is not attained by a path.

The matching polynomial closes the gap, unconditionally and with the sharp order of magnitude.

**Proposition 18** (matching bound). *Let  $F$  be a forest with matching number  $\nu \geq 1$  and let  $m_k$  be its number of  $k$ -matchings. Then every nonzero eigenvalue  $\theta$  of  $F$  satisfies  $\theta^2 \geq m_\nu/m_{\nu-1}$ .*

*Proof.* For a forest the characteristic polynomial is the matching polynomial, so  $\mu_F(x) = x^{n-2\nu}Q(x^2)$  with  $Q(y) = \sum_j (-1)^{\nu-j} m_{\nu-j} y^j$ , whose  $\nu$  roots  $y_1, \dots, y_\nu$  are the squares of the nonzero eigenvalues and are positive. Reading off the bottom two coefficients,  $e_\nu(y) = m_\nu$  and  $e_{\nu-1}(y) = m_{\nu-1}$ , so  $\sum_j y_j^{-1} = m_{\nu-1}/m_\nu$ ; every term of a sum of positive numbers is at most the sum.  $\square$

Write  $R(F) = m_{\nu-1}/m_\nu = \sum_j \theta_j^{-2}$ , the sum over the  $\nu$  positive eigenvalues. The key point is that deleting an exposed vertex cannot decrease  $R$ , and this is interlacing rather than combinatorics.

**Lemma 19.** *Let  $F$  be a forest with matching number  $\nu$  and let  $u$  be a vertex with  $\nu(F-u) = \nu$ . Then  $R(F) \leq R(F-u)$ .*

*Proof.* A forest has exactly  $2\nu$  nonzero eigenvalues, so  $F$  and  $F-u$  each have exactly  $\nu$  positive ones,  $\theta_1 \geq \dots \geq \theta_\nu > 0$  and  $\theta'_1 \geq \dots \geq \theta'_\nu > 0$ . Cauchy interlacing for the deletion of a vertex gives  $\theta'_j \leq \theta_j$  for every  $j$ , and all are positive, so  $\theta_j^{-2} \leq (\theta'_j)^{-2}$ . Sum over  $j$ .  $\square$

**Theorem 20.** *Every forest  $F$  with matching number  $\nu \geq 1$  satisfies  $R(F) = m_{\nu-1}/m_\nu \leq \binom{\nu+1}{2}$ , with equality if and only if  $F$  is  $P_{2\nu}$  together with isolated vertices. Consequently every nonzero eigenvalue of  $F$  satisfies*

$$\theta^2 \geq \frac{2}{\nu(\nu+1)}.$$

*Proof.* If  $|V(F)| > 2\nu$  some vertex  $u$  is missed by a maximum matching, and that matching survives in  $F-u$ , so  $\nu(F-u) = \nu$  and Lemma 19 applies. Iterating reaches a forest  $F^*$  on  $2\nu$  vertices with matching number  $\nu$ , that is with a perfect matching, and  $R(F) \leq R(F^*)$ . A forest has at most one perfect matching, since the symmetric difference of two would contain a cycle, so  $m_\nu(F^*) = 1$  and  $R(F^*) = m_{\nu-1}(F^*)$ . By the classical theorem of Gutman that the path is the unique maximizer of  $m_k$ , for each fixed  $k$ , among forests on a given number of vertices,  $m_{\nu-1}(F^*) \leq m_{\nu-1}(P_{2\nu}) = \binom{\nu+1}{\nu-1} = \binom{\nu+1}{2}$ . The last claim is Proposition 18 with  $R \leq \nu(\nu+1)/2$ .  $\square$

Since a  $\theta$ -critical tree of order  $n$  has  $\nu \leq n/2$ , Theorem 20 inverts to a lower bound on  $\kappa$ , and hence to a bound with no free parameter left in it.

**Theorem 21.** *Let  $G$  have minimum degree  $\delta \geq 3$  and maximum degree  $\Delta$ , and let  $\theta \neq 0$  be an eigenvalue of  $T_G$ . Then*

$$\theta^2 > \frac{8(\delta-2)^2}{(\Delta-2)(\Delta+2\delta-6)}, \quad \text{so in particular} \quad |\theta| > \frac{2\sqrt{2}(\delta-2)}{\Delta}(1+o(1)).$$

*Proof.* By Lemma 4.2 of [14] each component of a  $\theta$ -Aomoto subset is a  $\theta$ -critical tree, so by Theorem 20 its matching number satisfies  $\nu(\nu+1) \geq 2/\theta^2$  and its order is at least  $2\nu \geq \sqrt{1+8/\theta^2} - 1 =: K$ . Hence  $\kappa(\theta) \geq K$  and Proposition 16 gives  $\Delta > (\delta-2)K + 2$ . Solving for  $\theta^2$ , and using  $(\Delta + \delta - 4)^2 - (\delta - 2)^2 = (\Delta - 2)(\Delta + 2\delta - 6)$ , gives the stated bound.  $\square$

This is the structural statement the subject was missing. A nonzero point spectrum is not merely a matter of some degree spread being present; the spread must grow like  $1/|\theta|$ , so

eigenvalues near zero are expensive. It also explains the shape of every known counterexample to Conjecture 5: Hall's has  $\delta = 1$ , for which the bound is vacuous.

Proposition 16 is machine-checked in `RamaLean/DegreeBoundKappa.lean` as `degree_bound_kappa`, with `degree_bound_of_kappa_two` recovering Proposition 13 and `no_eigenvalue_of_spread_lt_kappa` the contrapositive. Proposition 18, the summation step of Lemma 19, and the final inequality of Theorem 20 are machine-checked in `RamaLean/MatchingEigenvalueBound.lean` as `eigenvalue_sq_ge_matching_ratio`, `sum_inv_sq_antitone` and `eigenvalue_sq_ge_two_div`. As throughout, the criterion of [14], Cauchy interlacing and Gutman's theorem are cited rather than reproved, and enter the Lean statements as hypotheses.

## 5.8 The degree bound is sharp

Propositions 13 and 16 are not merely true but attained, and the witness is small and explicit. At  $\delta = 3$  and  $\theta = \pm 1$ , whose  $\theta$ -critical tree is  $K_2$  and so  $\kappa = 2$ , both read  $\Delta \geq 5$ .

The bound of Theorem 21 is attained. Three constructions do it, one for each of the pairs  $(\delta, \kappa) = (3, 2)$ ,  $(3, 3)$  and  $(4, 2)$ : take  $c$  disjoint copies of the  $\theta$ -critical tree  $B$  together with  $b$  boundary vertices, and give every branch vertex enough boundary neighbours to reach degree  $\delta$ , sharing the resulting edges out equally. The counting chain fixes every parameter, since  $e = ((\delta - 2)|B| + 2)c \leq \Delta b$  and  $b \leq c - 1$  force  $c$ , then  $b$ , then  $n$ .

| $\delta$ | $B$   | $\kappa$ | $\theta$      | bound           | $\Delta$ | $c, b$ | $n$ |
|----------|-------|----------|---------------|-----------------|----------|--------|-----|
| 3        | $K_2$ | 2        | $\pm 1$       | $\Delta \geq 5$ | 5        | 5, 4   | 14  |
| 3        | $P_3$ | 3        | $\pm\sqrt{2}$ | $\Delta \geq 6$ | 6        | 6, 5   | 23  |
| 4        | $K_2$ | 2        | $\pm 1$       | $\Delta \geq 7$ | 7        | 7, 6   | 20  |

In each row  $\Delta$  equals the least value the bound permits, and  $n$  is the least possible for that  $\Delta$ . These are not three coincidences: the chain forces the construction, so the same argument runs for every pair.

**Theorem 22** (sharpness). *Let  $\delta \geq 3$  and let  $B$  be a  $\theta$ -critical tree of order  $\kappa \geq 2$ , and put  $W = (\delta - 2)\kappa + 2$ . There is a graph  $G$  with  $\delta(G) = \delta$ ,  $\Delta(G) = W + 1$  and  $\theta \in \text{Spec}_p T_G$ , on  $\kappa(W + 1) + W$  vertices. Both  $\Delta$  and the order are the least possible. In particular Proposition 16 is sharp for every such  $(\delta, \kappa)$ , and since  $P_\kappa$  is  $\theta$ -critical at  $\theta = 2 \cos \frac{\pi}{\kappa+1}$ , for every  $\kappa \geq 2$ .*

*Proof.* Take  $c = W + 1$  copies of  $B$  and  $b = W$  boundary vertices, and give each branch vertex  $v$  exactly  $\delta - \deg_B(v)$  boundary neighbours. The number of cross edges is  $\sum_v (\delta - \deg_B v) = c(\delta\kappa - 2(\kappa - 1)) = Wc = W(W + 1) = (W + 1)b$ , so the degrees can be shared out with every boundary vertex receiving exactly  $W + 1$ ; this is a realizable simple bipartite degree sequence because each  $\delta - \deg_B v \leq \delta - 1 \leq W = b$ . Branch vertices then have degree  $\delta$  and boundary vertices  $W + 1 \geq \delta$ , so  $\delta(G) = \delta$  and  $\Delta(G) = W + 1$ . The branch set  $S$  induces  $c$  disjoint copies of  $B$ , a forest each of whose components has  $\theta$  as an eigenvalue, and  $|\partial_G S| = b = c - 1 < c$ , so  $S$  is a  $\theta$ -Aomoto subset and [14] gives  $\theta \in \text{Spec}_p T_G$ . Minimality: the chain of Proposition 16 reads  $Wc \leq \Delta(c - 1)$ , so  $\Delta = W + 1$  forces  $c \geq W + 1$ , then  $e = Wc \leq \Delta b$  gives  $b \geq W$ , while  $b \leq c - 1$ ; hence  $c = W + 1$ ,  $b = W$  and  $|V| \geq \kappa(W + 1) + W$ .  $\square$

The first row is the smallest instance of all and is worth stating separately, with its point spectrum computed outright.

**Proposition 23.** *Let  $G$  be the graph on 14 vertices consisting of five disjoint edges together with four further vertices, where each of the ten edge-endpoints is joined to two of the four, distributed so that each of the four receives five such edges. Then  $\delta(G) = 3$ ,  $\Delta(G) = 5$ , and the point spectrum of  $T_G$  is exactly  $\{+1, -1\}$ . Moreover no graph with  $\delta \geq 3$  and a nonzero eigenvalue of its universal cover has  $\Delta < 5$ , or has fewer than 14 vertices when  $\Delta = 5$ .*

*Proof.* The five disjoint edges form the set  $S$ :  $G[S]$  is a forest with  $\text{cc} = 5$  components, each having  $\pm 1$  as an eigenvalue, and  $|\partial_G S| = 4 < 5$ , so  $S$  is a  $(\pm 1)$ -Aomoto subset and [14] gives  $\pm 1 \in \text{Spec}_p T_G$ . That the point spectrum is no larger is the computation below. For minimality, the chain in the proof of Proposition 16 with  $\delta = 3$ ,  $\kappa = 2$  reads  $4c \leq \Delta(c - 1)$ , so  $\Delta = 5$  forces  $c \geq 5$ ; then  $e = 4c = 20 \leq \Delta b = 5b$  gives  $b \geq 4$ , while the Aomoto inequality gives  $b \leq c - 1 = 4$ . Hence  $b = 4$ ,  $c = 5$ , and  $|V| \geq 2 \cdot 5 + 4 = 14$ .  $\square$

The exact certificate of Remark 10 identifies the point spectrum outright, since  $\theta$  is an eigenvalue exactly when it is a root of every  $\mu_{G \setminus \Gamma}$ , so  $\text{Spec}_p T_G$  is the root set of the integer polynomial  $E_G = \gcd_{\Gamma} \mu_{G \setminus \Gamma}$ . For this graph there are 504 two-regular subgraphs and

$$\mu_G = (x^2 - 1)(x^{12} - 24x^{10} + 206x^8 - 756x^6 + 1081x^4 - 364x^2 + 16), \quad E_G = x^2 - 1.$$

So  $\Delta = 5 = 2\delta - 1$  is attained, and both propositions are sharp at  $\delta = 3$ . For contrast,  $E_G = x^4(x^2 - 3)$  for the 14-vertex construction of [18], which has  $\delta = 3$  but  $\Delta = 9$ .

Sharpness should not be read as encouragement. The bound is attained by a graph built to attain it, and in 397 random graphs with  $\delta \geq 3$  on at most 12 vertices, sampled across a range of densities,  $E_G$  had no nonzero root even once. A nonzero point spectrum is not a matter of degrees being large enough; it needs many components with a small boundary, which is a sparse and highly structured condition that random graphs do not meet.

## 5.9 An exact search in the trap-free class

Corollary 14 identifies a class in which a counterexample would be unconditional, which makes it worth searching directly. Among trap-free graphs the  $(3, 4)$ -biregular ones are the natural target, for a reason that is not obvious in advance: a random graph with degrees in  $\{3, 4\}$  has no spectral gap at all, its cover spectrum running continuously through the origin, so there is nowhere for a root to hide. The biregular case is the exception. There

$$\text{spec}(T) = \{0\} \cup \pm[\sqrt{3} - \sqrt{2}, \sqrt{3} + \sqrt{2}] = \{0\} \cup \pm[0.317837\dots, 3.146264\dots],$$

and the complement has two windows. Both must be checked: Heilmann–Lieb confines the zeros of  $\mu_G$  only to  $|x| < 2\sqrt{3} = 3.4641\dots$ , which exceeds  $\sqrt{3} + \sqrt{2}$ , so the upper window is not empty on those grounds.

The search is exact, with no floating point in the verdict. Writing  $\mu_G = x^{n-2\nu}Q(x^2)$ , the polynomial  $Q$  is hyperbolic with all roots real and strictly positive, so its Budan–Fourier count is exact rather than an upper bound with even defect, and it requires no polynomial remainder sequence and hence suffers none of the coefficient growth of a Sturm chain. Matching counts are computed by a subset dynamic program over the smaller side of the bipartition and  $Q$

is evaluated in exact integer arithmetic at rational endpoints chosen to lie strictly inside the window being tested. Each graph additionally carries the free consistency check  $V(0) = \deg Q$ , and a graph failing it is counted as undecided rather than passed.

The sweep (`code/rust_biregular`) tested 82896 graphs: exhaustively at  $t = 1$  and  $t = 2$ , meaning all 2895 connected labelled  $(3, 4)$ -biregular graphs on 14 vertices, and 20000 random graphs at each of  $t = 3, \dots, 6$ , up to  $n = 42$ . No zero of  $\mu_G$  lies outside  $\text{spec}(T_G)$  anywhere in the sample, and no graph failed the consistency check.

We state plainly what this is and is not. The individual verdicts are exact; the coverage is not exhaustive beyond  $n = 14$ , the random graphs are drawn by stub pairing with rejection and so are not uniform, and the run terminated because the range of  $t$  was exhausted rather than because a time budget expired. It is evidence for Conjecture 5 in the trap-free class, not a proof. Its value is that it is a negative result obtained where a positive one would have been unconditional: the one region of the class with a spectral gap, and the region in which the Aomoto escape is machine-checked to be unavailable, contains no counterexample up to 42 vertices.

## 5.10 The width threshold, and what it costs

The searches reported here and in [18] filter the detected gaps of  $\text{spec}(T)$  by a minimum width before looking for roots inside them, and skip a graph when no gap survives. The threshold used throughout is 0.05. Hall’s own counterexample does not survive it:  $\text{spec}(T)$  for his graph has a gap

$$(2.220, 2.260), \quad \text{width } 0.040, \quad \sqrt{5} = 2.23607 \dots \in (2.220, 2.260),$$

and  $0.040 < 0.05$ , so his graph is scored as having no gap and reported clean (`code/gapfilter_blindspot.py`). The detector finds the gap; the threshold discards it.

Two measurements fix how much this costs, and they point in opposite directions.

Re-running the minimum-degree-three families at 0.015 changes nothing. Exactly 87 of the graphs swept present a gap at 0.05 and exactly 87 at 0.015, and no root in any of them is a violation (`code/rerun_lowgap.py`). For those families the threshold was not discarding search space, and the recorded verdicts stand unweakened. We had expected otherwise and record the miss.

But the threshold does exclude the width regime in which counterexamples actually occur. A search of one-cut constructions run at 0.015, with the exact eigenvalue test of §5.4 in place of a numerical one, returns four counterexamples to Conjecture 5 on 36 and 37 vertices, with minimal polynomials  $x^6 - 12x^4 + 25x^2 - 7$ ,  $x^6 - 13x^4 + 26x^2 - 7$ ,  $x^6 - 13x^4 + 32x^2 - 13$  and  $x^6 - 13x^4 + 32x^2 - 12$ , decays 0.94 to 0.98, and gap widths of 0.040 in every case (`data/lowgap_counterexamples.json`). Together with Hall’s, that is five known counterexamples, all of them in gaps of width 0.040, which is the smallest a scan of step 0.02 can resolve, and all of them below the threshold every search here used.

How wide are those gaps really? The scan reports 0.040 in all five cases, which is two of its 0.02 steps and therefore its resolution floor rather than a measurement. Resolving them needs a detector that returns contiguous intervals, which `code/gap_scale2.py` supplies: it replaces the real fixed-point iteration by the complex cavity recursion at  $z = \lambda + i\eta$ , which converges at every  $\lambda$ , and separates band from gap from atom by the scaling of  $|\text{Im} \sum_w G_w|$  between two values of  $\eta$  a factor 100 apart. It reproduces the exact biregular gap  $(0, |s - t|)$

for  $K_{3,4}$  and  $K_{2,3}$  and reports no gap for a 3-regular graph. Scanned at step 0.002, the gaps holding the five violating roots have widths

$$0.030, \quad 0.002, \quad 0.018, \quad 0.026, \quad 0.038,$$

every one below the 0.05 threshold that discarded them. The phenomenon really does live in gaps narrower than the filter, and the filter really does exclude every example known to us.

Two things we got wrong on the way to that are worth recording, since both are easy traps. The non-convergence of the real iteration is not a defect: it corresponds to  $\lambda$  lying in a band, and scoring it as spectrum is correct. And a fine scan that takes the extremes of the certified points merges disjoint gaps across an intervening band; doing that produced an apparent gap of width 0.1035 where there are really two gaps of 0.04 and 0.025 with a band between them (`code/gapwidth_underreport.py`, retracted in place).

The remedy for a search is therefore a finer step and a much lower cutoff, not a different solver. At step 0.02 a gap of width 0.002 cannot be seen at all.

The violating root sits at 40 to 61 percent of the way across its gap in four of the five cases and at 99 percent in the fifth, which is accordingly the least robust of the set.

One of the four, the 36-vertex example, carries a certified band exclusion as well. Its automorphism group has order 5040 and its 84 directed edges fall into 12 orbits of size 7, so the Angel–Friedman–Hoory system has twelve unknowns; solved at 60 digits it gives a Collatz–Wielandt constant  $c = 0.9709248\dots$  against a solution uncertainty of  $5 \times 10^{-58}$ , a margin of 55 orders of magnitude (`code/exact_certificate36.py`). That is short of Hall’s certificate, which is exact in  $\mathbb{Q}(\sqrt{5}, \sqrt{41})$ , and the shortfall is not removable by more effort: two of the twelve ratios lie in  $\mathbb{Q}(\theta)$  with closed forms, and the other ten admit no polynomial relation of degree at most 30 with coefficients below  $10^{20}$  at 200 digits. A large automorphism group does not give a small ratio field. We record also that integer relation search returned plausible expressions in  $\mathbb{Q}(\theta)$  for all twelve, of which ten were spurious and were caught only by exact reduction in  $\mathbb{Q}[x]/(f)$ .

The four new examples have minimum degree one, so they bear on Conjecture 5 and not on Conjectures 8 or 9, which remain open. Re-running the minimum-degree-three families at the lower threshold changes nothing: 87 graphs present a gap at either setting and no root in any of them is a violation (`code/rerun_lowgap.py`), so those recorded verdicts stand unweakened.

## 5.11 Wheels saturate the bound

Wheels are the one family of minimum degree three here that carries genuine spectral gaps, so they are where a counterexample to Conjecture 8 would most plausibly appear. Re-checking them with the corrected detector of §5.10 and the exact eigenvalue test, and extending the range, none of  $W_5, \dots, W_{50}$  has a root of  $\mu$  inside a gap. The roots straddle the gap: the bulk lie below its lower edge, near 2, and the largest lies above its upper edge.

What is new is the rate at which they do so. Writing

$$\text{margin}(m) = (\text{largest root of } \mu_{W_m}) - (\text{top of the gap below it}),$$

the margin falls monotonically from 0.637 at  $m = 5$  to 0.194 at  $m = 50$ , and a log-log fit over that range gives

$$\text{margin}(m) \approx 1.463 m^{-0.5125}, \quad R^2 = 0.999,$$

so the family saturates the covering bound asymptotically without violating it (`code/wheel_saturation.py`). The computation uses the closed form  $\mu_{W_m} = x(\mu_{P_m} - \mu_{P_{m-2}}) - m\mu_{P_{m-1}}$ , checked against the deletion recursion before use.

This is the same phenomenon the companion paper records for the biregular family, where the margin decays like  $n^{-2/3}$ , the soft-edge exponent [17]. That figure has been replicated here independently, with matchings recounted by a subset dynamic program on independently generated graphs: the fit gives  $n^{-0.6629}$  at  $R^2 = 0.9996$ , reproducing the published series to four decimals at  $n = 12$  (`code/softedge_audit.py`). It is worth noting that the biregular claim needs no spectral instrument at all, the band edge for a biregular cover being the closed form  $|\sqrt{q-1} - \sqrt{d-1}|$ , so none of the failures recorded in §5.10 can touch it.

So two families saturate, at  $-0.51$  and  $-0.66$ , and the exponents are genuinely different. We do not read that as two edge universality classes. The biregular family has bounded degree,  $d$  and  $q$  fixed as  $n$  grows, and the wheel family does not, since the hub of  $W_m$  has degree  $m$ . Under the soft-edge heuristic a margin  $N^{-1/(1+\alpha)}$  corresponds to a density vanishing like  $(x - \tau)^\alpha$ , so  $-2/3$  is the square-root edge  $\alpha = \frac{1}{2}$  and  $-\frac{1}{2}$  would be a linear edge  $\alpha = 1$ ; but an unbounded degree could produce the same shift for unrelated reasons, and nothing here separates the two explanations.

What is unambiguous is the part that bears on Conjecture 8: both families saturate, the margin tends to zero in each, and an argument yielding a margin bounded below independently of size cannot prove the conjecture on either.

## 5.12 What this leaves

Conjecture 5 is false, by Hall’s counterexample. Conjectures 8 and 9 are open, and Proposition 12 says that the branch mechanism cannot settle them in the negative: it manufactures eigenvalues of the cover, not counterexamples.

That is a sharper statement than it may look. Any counterexample must have its offending root avoid every  $\theta$ -Aomoto subset, so a search should be organised around the Aomoto inequality rather than around degrees or connectivity. Hall’s graph satisfies this by giving each branch a private boundary of two vertices; we know of no systematic way to do that at minimum degree three.

The biregular case, which survives the refutation of Conjecture 5 and is where the phenomenon lives, is [17].

## 6 Related work and formalization

The  $d$ -matching polynomial and the interval  $[-\rho, \rho]$  are due to Hall, Puder and Sawin [2], whose Question 6.3 asks for exactly the strengthening considered here; Heilmann and Lieb [6] give the classical interval, Godsil [7, 8] the path-tree method, and Angel, Friedman and Hoory [13] the ratio criterion the certificates use. The cycle identity behind Theorem 21 was proved combinatorially by Cochran, Groothuis, Herring, Rohatgi and Stucky [3]; our route is independent and elementary. Song, Fan and Miao [5] pose the biregular case, which is [17].

The Lean 4/Mathlib sources are in `RamaLean/` of the accompanying repository and carry no `sorry` and no custom axioms: the closed form at first Betti number one, the counterexample’s algebra, and both halves of Lemma 7. The standing gap is that Mathlib has no matching

polynomial, so the expansion in Lemma 7 and the Angel–Friedman–Hoory criterion are cited rather than derived.

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